EXCEL
WITH
AI

Imagine your manager gives you an Excel file and says:

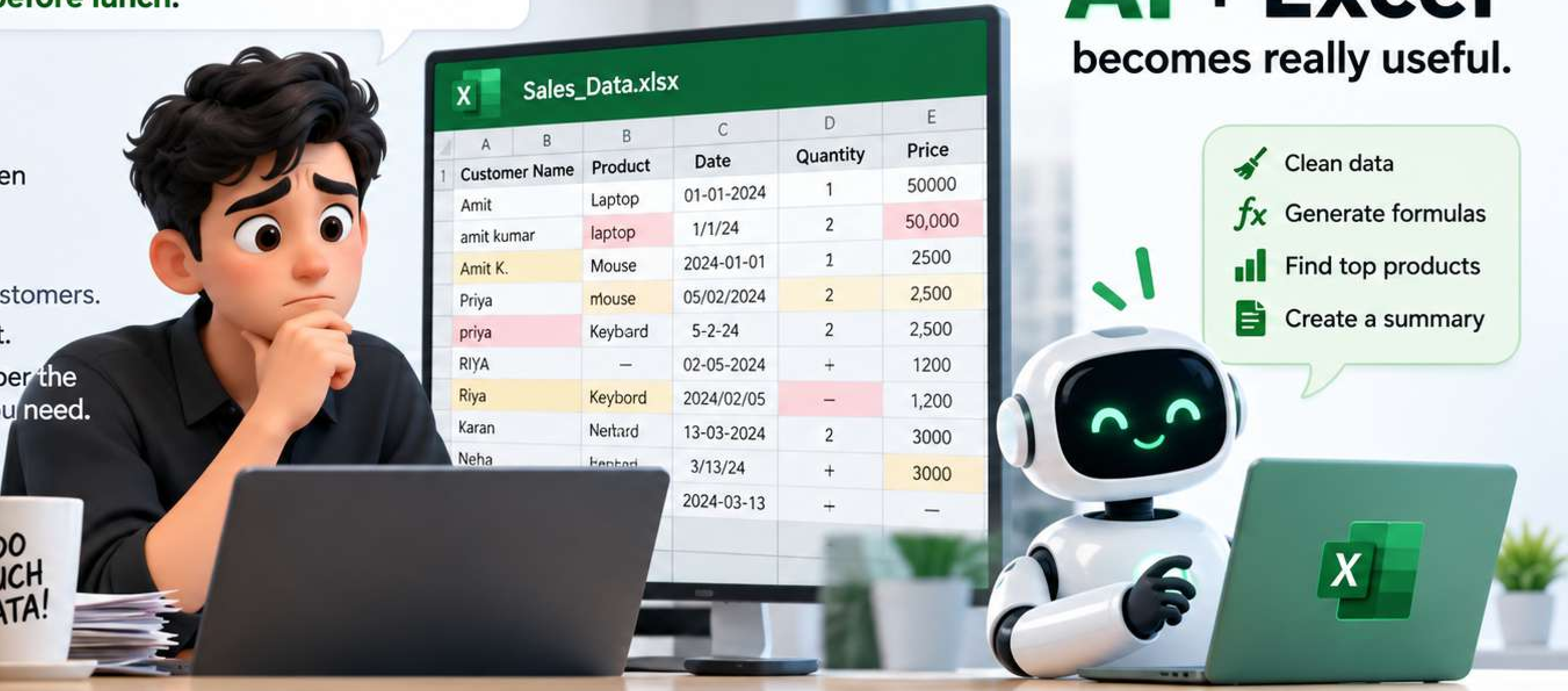
"Clean this data, calculate the sales totals, find the top-performing products, and give me a summary before lunch."

You open the file.

There are **2,000 rows**.

- Some names are written differently.
- Some cells are empty.
- There are duplicate customers.
- Dates are inconsistent.
- And you don't remember the exact Excel formula you need.

What would you do?



This is where

AI + Excel

becomes really useful.

- Clean data
- Generate formulas
- Find top products
- Create a summary

But there's one rule for today's class:

AI can help you write a formula, but you must understand what that formula is doing.

Let's Start With a Simple Example

Excel with AI

1 Your Excel Data

Imagine your Excel sheet contains:

	A	B
1	**Student**	**Marks**
2	Aman	85
3	Riya	72
4	Rahul	91
5	Neha	64

Your teacher asks:

"Calculate the average marks."

2 Ask AI for the Formula

You don't remember the formula, so you ask:

What Excel formula calculates the average of cells B2 to B5?

The Excel formula is:

=AVERAGE(B2:B5)

3 Ask AI to Explain

Now ask:

Explain this formula in simple language.

The **AVERAGE** function calculates the mean value of the selected cells.

In this case, it adds the values in cells B2 to B5 and divides by the number of cells.

Key Takeaway:

AI can give you the formula, but you must understand what it is doing.

1. Natural-Language Excel Prompts

You can describe what you want in **normal** language. AI helps translate it into **Excel** logic.

1 Example: Total Sales

Give me a formula to calculate the total sales in column D.

Use this formula:

=SUM(D2:D100)

	C	D
1	**Product**	**Sales**
2	Laptop	50,000
3	Mouse	10,000
4	Keyboard	8,000
5	Monitor	25,000
...	...	...

2 Example: Calculate Profit

I want to calculate profit by subtracting cost from selling price.

Use this formula:

=B2-C2

	A	B	C	D
1	**Product**	**Selling Price**	**Cost**	**Profit**
2	Laptop	60,000	40,000	=B2-C2
3	Mouse	1,500	800	=B3-C3
4	Keyboard	2,000	1,200	=B4-C4
5	Monitor	15,000	10,000	=B5-C5
...	...	...	...	...

Key Takeaway:

You describe the business problem.

AI helps translate it into Excel logic.

*"You don't need to remember every formula. You just need to know **what you want to achieve**."*

Let's Think Like a BBA Student

Suppose you're working as a sales intern. Your Excel sheet looks like this:

	A	B	C	D
1	**Product**	**Units Sold**	**Price**	**Revenue**
2	Shoes	20	₹2,000	
3	Bags	15	₹1,500	
4	Watches	10	₹3,000	

1 Identify the Business Logic

Revenue is calculated as:

Revenue = Units Sold × Price

2 Use the Excel Formula

In Excel, for the first row (Shoes):

=B2*C2

Then copy the formula down for the other products.

3 Ask AI to Explain

Prompt: "Explain why B2*C2 calculates revenue."

B2 contains the number of units sold and C2 contains the price per unit. Multiplying them (B2*C2) gives the total revenue for that product.

Key Takeaway:

You're not just using a formula. You're applying business logic.

Think.
Analyze.
Apply.

Don't Become a Formula Copier

AI is a powerful helper, but you must understand what the formula is doing.

Understand
Think
Learn
Don't just copy!

The Common Problem

Student: "Give me the formula.

AI: =IFERROR((B2-C2)/B2,0)

Teacher: What does IFERROR do?

Student: "AI gave me the formula.

This is exactly what we don't want.

A Better Approach

Instead of just copying, ask AI to explain.

Prompt:
"Explain this formula step by step in simple language.
=IFERROR((B2-C2)/B2,0)"

You can also ask:
"Break this formula into smaller parts."

Example Explanation

=IFERROR((B2-C2)/B2,0)

1. **(B2-C2)**
   Subtracts C2 from B2.
2. **(B2-C2)/B2**
   Divides the result by B2 (e.g., to calculate a percentage).
3. **IFERROR(...,0)**
   If there is an error (like division by zero), return 0 instead of an error message.

So this formula calculates the percentage difference and shows 0 if there is an error.

Important Rule:

If you cannot explain a formula, don't blindly use it.

Understand.
Apply.
Grow.

Data Cleaning

In real business data, things are not always clean and organized.

Look at this customer data:

	A	B	C
1	**Name**	**City**	**Phone**
2	Aman	Kanpur	9876543210
3	aman	kanpur	9876543210
4	Riya	Delhi	9876543211
5	Rahul	Delhi	
6	Neha	Mumbai	9876543212

Real-world data is often messy!

What's wrong with this data?

Duplicate customer
Aman and **aman** are the same person.

Different capitalization
Kanpur and **kanpur** are written differently.

Missing phone number
Rahul's phone number is empty.

This is called **messy data.**

Key Takeaway:

Business data often has duplicates, inconsistencies, and missing values.
Data cleaning is an important step before analysis.

Clean Data
Better Insights
Smarter Decisions

Why Does Data Cleaning Matter?

Because bad data can lead to bad business decisions.

Clean Data
Better Decisions
Stronger Business

Customer Database (Messy Data)

Name	City	Phone
Aman	Kanpur	9876543210
aman	kanpur	9876543210
Riya	Delhi	9876543211
Rahul	Delhi	
Neha	Mumbai	9876543212

Aman and aman are the same customer, but they appear as two separate records.

Marketing Manager
"I want to send a promotional message to every customer."

What Can Go Wrong?

Send duplicate messages
The same customer receives multiple messages.

Count the same customer twice
Leads to inflated numbers.

Calculate incorrect sales
Data becomes unreliable.

Create inaccurate reports
Leads to poor business decisions.

Key Takeaway:

Bad data can lead to bad business decisions.

Clean Data
Reliable Insights
Better Results

Duplicate Removal

Excel can help identify and remove duplicates. But you must understand your data before deleting anything.

Don't just delete. Understand first!

Cleaner Data
Stronger Decisions

1 Example: Duplicate Records

	A	B
1	**Customer**	**Email**
2	Aman	aman@gmail.com
3	Aman	aman@gmail.com
4	Riya	riya@gmail.com

! The first two rows may represent the same customer.

2 Excel Tool: Remove Duplicates

Data | Review | View | Help

Text to Columns | Remove Duplicates | Data Validation | Consolidate

Data Tools

You can ask AI:
"How do I identify and remove duplicate rows in Excel based on email address?"

AI can give you step-by-step instructions with screenshots.

3 Important Question

Should you automatically delete every duplicate?

No. Because sometimes duplicate records are legitimate.

Example:
A customer may have:

- One order in January
- Another order in February

Their name and email will appear twice.
Those aren't duplicate transactions.
They're two legitimate purchases.

Important Rule:

Before removing duplicates, understand what one row represents.

Is one row a:

- Customer?
- Order?
- Transaction?
- Product?
- Visit?
- Something else?

Missing Values

In real data, some information is often missing. Look at this example:

	A	B	C
1	**Customer**	**Age**	**City**
2	Aman	20	Kanpur
3	Riya		Delhi
4	Rahul	21	
5	Neha	22	Mumbai

Different reasons need different solutions!

Why Is the Value Missing?

Customer didn't provide the information
Maybe the customer chose not to share their age.

Data wasn't collected
The field might not have been captured during data entry.

Technical error
There could have been a system or import error.

Doesn't apply
The information might not be relevant in some cases.

Common Mistake

"Fill every blank with something."

Don't do that.

Better Approach

1. Identify why the value is missing.
2. Understand the context.
3. Choose the right method.
4. Be consistent.

Key Takeaway:

Before filling missing values, understand why they are missing.

Better Data.
Better Analysis.
Better Decisions.

Can AI Fill Missing Values?

Yes, AI can suggest different ways to handle missing values, but you must use your judgment and understand the context.

AI gives options. You make the right decision!

Example: Missing Values in Age

	A	B	C
1	**Customer**	**Age**	**City**
2	Aman	20	Kanpur
3	Riya		Delhi
4	Rahul	21	
5	Neha	22	Mumbai

AI Might Suggest:

1. **Leave them blank**
   If the information is unknown and not required for analysis.
2. **Remove records**
   If missing values are few and not critical.
3. **Use a statistical value**
   Such as median or mean (only when appropriate).
4. **Investigate the original source**
   Check if the data can be collected again or corrected.

AI gives you options, but you need to choose the best approach based on your business problem.

Don't Do This!

Should you blindly fill every missing age with 20?

No. That would create fake data and lead to incorrect analysis.

Important Rule

Never confuse an estimated value with an actual value.

If you estimate something, it should be clearly understood as an estimate.

Be Transparent

If you fill missing values using an estimate (e.g., median), mention it in your report.

Key Takeaway:

AI can help you handle missing values, but use your judgment.

Better Data.
Better Analysis.
Better Decisions.

Formatting Problems

Messy spreadsheets often have formatting problems. The same type of information can appear in different formats.

Same data. Different formats!

Clean Formats Clear Insights

Example: Different Date Formats

	A
1	**Date**
2	12/01/26
3	January 15, 2026
4	2026-01-20
5	21 Jan 26

These may represent the same type of information, but they are formatted differently.

How AI Can Help

"How can I standardize different date formats in Excel?"

AI may suggest:

1. Use Excel's Date format option
2. Use the TEXT or DATEVALUE function
3. Convert text to date using "Text to Columns'
4. Apply a consistent date format (e.g., dd-mmm-yyyy)

Important Tips

Don't assume the transformation is always correct.
Check a few rows manually.

Understand the format you want.
E.g., 20-Jan-2026 or 2026-01-20

Be consistent.
Use the same date format for the entire column.

Watch for invalid dates.
Some entries may not convert correctly.

Key Takeaway:

AI can help you identify and fix formatting problems, but always verify the results.

Cleaner Data
Fewer Errors
Better Analysis

Microsoft Copilot in Excel

Use natural language to work with your spreadsheets.

Talk to your data
Get insights faster
Work smarter!

Same Excel.
A Smarter You.

What Can You Ask?

- "Summarize the sales data."
- "Identify the products with the highest sales."
- "Create a formula to calculate profit margin."
- "Highlight unusual values."
- "Show me the monthly sales trend."
- "Create a chart of sales by product category."

File | Home | Insert | Draw | Page Layout | Formulas | Data | Review | View | Help

A1

	A	B	C	D
1	**Product**	**Sales**	**Cost**	**Profit**
2	Shoes	₹40,000	₹25,000	₹15,000
3	Bags	₹30,000	₹18,000	₹12,000
4	Watches	₹75,000	₹45,000	₹30,000
5	Belts	₹20,000	₹12,000	₹8,000
5	Sunglasses	₹35,000	₹20,000	₹15,000

Sales by Product

Key Insights (Example)

- Watches have the highest sales.
- Total sales: ₹2,00,000
- Average profit margin: 37.5%
- No major unusual values found.

Copilot

Summarize the sales data.

Here's a summary of your sales data:

- Total sales: ₹2,00,000
- Top product: Watches (₹75,000)
- Average sales per product: ₹40,000

Insert summary table

Show me the monthly sales trend.

Create a formula for profit margin.

Highlight unusual values.

Ask Copilot anything...

Why Use Copilot?

Saves time
Get answers quickly.

Simplifies complex tasks
No need to remember formulas or menu steps.

Helps with analysis
Find trends, insights and outliers.

Easy to use
Just type what you need in simple language.

Focus on business questions
Spend more time thinking, less time clicking.

Key Takeaway:

You can communicate with your spreadsheet using natural language.

Better Insights
Higher Productivity
Smarter Decisions

But AI Doesn’t Understand Your Business Automatically

AI can work with your data, but it doesn’t know your business definitions unless you tell it.

Be clear.
Define your terms.
Get better results!

Your Business.
Your Definition.
Better Answers.

Example: Who is the Best Customer?

	A	B
1	**Customer**	**Purchase**
2	Aman	₹5,000
3	Riya	₹10,000
4	Rahul	₹7,000

You Ask AI:

“Find the best customer.”

AI might assume:

Riya is the best customer because she has the highest purchase value (₹10,000).

But What Does “Best Customer” Mean?

It could mean different things depending on your business:

- Highest total spending
- Most frequent buyer
- Highest profit
- Longest relationship
- Highest lifetime value

Key Point

The spreadsheet **doesn’t know** **your business definition** unless you tell it.

Important Rule

AI can calculate your **definition.**
It cannot automatically know your **definition.**

Key Takeaway:

Be clear about your business question, define it, and then use AI to get the right answer.

Better Questions.
Better Analysis.
Better Decisions.

Main Activity — Clean a Messy Dataset (Part 1)

You are given a messy customer dataset. Your job is to clean it using Excel (with or without AI).

Same data.
Cleaner insights.
Better decisions!

Messy Customer Dataset

	Customer ID	Name	Email	City	Date	Sales (₹)	Cost (₹)
1	C001	Aman	aman@gmail.com	Kanpur	12/01/26	5,000	3,000
2	C001	aman	AMAN@gmail.com	kanpur	January 12, 2026	5000	3000
3	C002	Riya	riya@gmail.com	Delhi	2026-01-15	10,000	6,500
4	C003	Rahul	rahul@gmail.com	Delhi	15 Jan 26		4,000
5	C004	Neha	neha@gmail.com	Mumbai	01-20-2026	8,000	5,000
6	C005	Priya	priya@gmail.com	mumbai	20/01/26	7500	4,500
7	C006	Sohan	sohan@gmail.com	Pune	January 25, 2026	12,000	8,000
8	C006	Sohan	sohan@gmail.com	Pune	25-01-2026	12,000	8,000
9	C007	Anjali	anjali@ gmail.com	Delhi	02/05/26	9,000	
10	C008	Karan	KARAN@gmail.com	kanpur	5 Feb 2026	7000	4500
11	C009	Meera	meera@gmail.com	Bangalore	2026/02/10	11,000	7,000
12	C010	Vikram	vikram@gmail.com	Bengaluru	10-Feb-26		6,000

Your Tasks

1. Remove duplicate rows
2. Handle missing values (decide the best approach)
3. Fix inconsistent capitalization
4. Standardize date formats
5. Remove extra spaces
6. Correct formatting (numbers, text, etc.)
7. Ensure sales and cost values are clean and usable

Things to Keep in Mind

- Understand what one row represents.
- Don't blindly delete every duplicate.
- Don't fill missing values with random data.
- Standardize formats consistently.
- Always verify the cleaned data.

Goal

Create a **clean, accurate, and consistent** customer dataset that is ready for analysis.

Pro Tip

You can ask AI for guidance, formulas, or steps, but **always think and verify the results yourself.**

Main Activity — Clean a Messy Dataset (Part 2)

Follow the steps below to clean the dataset, identify issues, and create useful formulas.

Clean Data.
Better Insights.
Better Decisions!

1 Understand the Dataset

Before touching anything, ask:

"What does each row represent?"

For example:
One row = one customer transaction.

Your cleaning decisions depend on this.

Know your data.
Clean with a purpose.

2 Find the Problems

Ask AI for a data-cleaning checklist:

"I have an Excel dataset containing customer transaction data. Help me create a data-cleaning checklist. Look for potential issues such as duplicate records, missing values, inconsistent text formatting, date formatting problems, invalid values, extra spaces, and incorrect data types. For each issue, explain how I can verify it in Excel. Do not modify or invent any data."

Then inspect the dataset yourself.

Common Issues to Check

- Duplicate records
- Missing values
- Inconsistent text formatting
- Date formatting problems
- Invalid values
- Extra spaces
- Incorrect data types

3 Clean the Data

Based on your inspection, clean the data.

Duplicate records	Check if they are truly duplicates before removing.
Missing values	Decide: keep blank, correct from source, or handle statistically.
Text formatting	Standardize capitalization (e.g., kanpur → Kanpur).
Extra spaces	Use functions like **=TRIM(A2)** (Remove leading/trailing spaces).
Invalid / incorrect formatting	Fix numbers, dates, text, etc.

4 Generate 3 Formulas Using AI

Create three useful formulas based on your cleaned dataset.

#	Purpose	Example Formula	What it Does
1	Total Revenue	=B2*C2	Multiplies quantity by unit price.
2	Profit	=D2-E2	Subtracts cost from revenue.
3	Profit Margin	=(D2-E2)/D2	Calculates profit as a percentage of revenue.

For each formula, ask AI:

"Explain this formula step by step and tell me what each cell reference represents."

Understand what you use.
Don't just copy formulas.

5 Test the Formula

Don't assume the formula is correct.

1. Take one row and calculate the answer manually.
2. Compare it with the Excel result.
3. If both match, good.
4. If not, investigate.

A formula producing a number **doesn't mean the number is correct.**

Key Takeaway:

Clean, accurate data + correct formulas = reliable analysis and better business decisions.

Clean Data
Confident Analysis
Real Impact

What Should Still Be Manually Verified?

AI can help find issues, but humans must verify and make the final decisions.

AI finds the issues. You understand the context.

Human Judgment Still Matters

1 Duplicate Records

Is it actually a duplicate or a legitimate repeated transaction?

Customer	Order Date
Aman	01-Jan-2026
Aman	15-Feb-2026

Same customer, different orders. These are not duplicates.

2 Missing Values

Why is the value missing?

- Customer didn't provide it?
- Not collected?
- Technical error?
- Not applicable?

Understand the reason before filling, removing, or estimating.

3 Outliers

Is ₹1,00,000 really an error or a genuine large purchase?

Customer	Sales (₹)
Riya	5,000
Karan	8,000
Meera	1,00,000

A high value may be valid (e.g., bulk order, festival sale). Investigate before taking action.

4 Incorrect Data

Some values are clearly wrong, but you must decide the correct value.

Name	Age
Rahul	28
Neha	250
Aman	22

Age = 250 is likely wrong, but you need to find the correct value from the source.

5 Business Meaning

AI sees numbers. You understand what they mean in the business context.

- Is the value reasonable for your industry?
- Was there a special event (e.g., Diwali sale)?
- Is it a one-time corporate order?
- Does it match business trends?

You decide what makes sense for your business.

Important Example

Your sales dataset shows:

Product A — Sales: ₹10,00,000

AI says: "This is an outlier. Remove it."

Would you remove it?

Not without investigation!

- Maybe Product A had a huge Diwali order.
- Maybe a corporate customer placed a bulk order.
- Maybe the number really is incorrect.

AI can identify an unusual value. But you decide whether it's an error.

Key Takeaway:

AI can find potential issues, but humans must verify, interpret, and make the final decisions.

Better Data
Better Insights
Better Business Decisions

The 5-Step Rule for AI + Excel

Use AI to work faster, but think critically and verify the results.

AI gives you options. You make the decisions.

Smarter Spreadsheets Brighter Decisions.

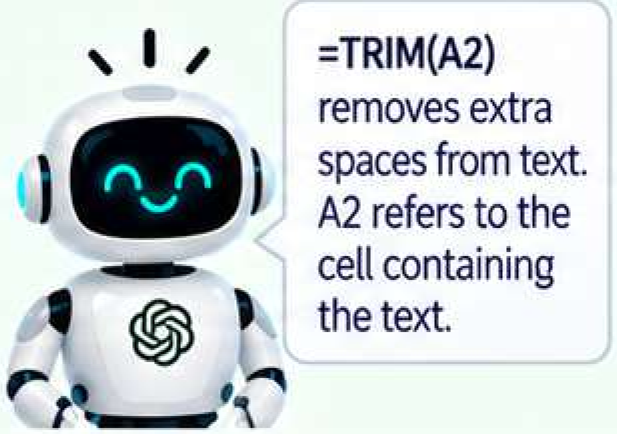
1 Ask

Tell AI what you want to do.

Be clear and specific about your task.

2 Understand

Ask AI to explain the formula or method.



Understand what it does and what each cell reference means.

3 Check

Test the result.

	A	B
1	**Original**	**After TRIM**
2	" Aman "	Aman
3	"Riya.	Riya

Try it on a few rows to see if it works as expected.

4 Verify

Compare against the original data.

Cross-check with the original data and, if needed, calculate manually.

5 Decide

Use your business judgment.

Consider the business context and decide the final action.

Key Takeaway:

AI helps you do the work. You ensure it's correct and meaningful.

Better Tools
Better Thinking
Better Results

Final Challenge

Don't just accept the result — understand, verify, and take informed decisions.

It's not just about the answer. It's about the process.

Better Data. Better Questions. Better Decisions.

Your Manager Says

"Clean this customer dataset and give me the total sales."

AI Does Everything

- Removes 200 duplicate rows
- Fills 50 missing values
- Changes 30 dates
- Creates a sales formula
- Gives you the final total

Total Sales: ₹ 48,75,000

Before Sharing the Result, Ask:

1. What did AI change?
2. Why did it change it?
3. Did it make any assumptions?
4. Are the formulas correct?
5. Did the cleaning change the business meaning of the data?

Don't Just Send the Result

- You might miss important errors.
- You might accept incorrect assumptions.
- You might share misleading information.
- You could affect business decisions.

> *"Using AI intelligently means understanding the changes, not just accepting the output."*

Be a Responsible AI User

- Review the changes made by AI.
- Understand the reasoning.
- Verify key values and formulas.
- Use your business judgment.
- Share the result only when you are confident.

Key Takeaway:

AI can process data. You ensure it's correct, meaningful, and ready for decisions.

Same Tools.
Better Verification.
Greater Impact.